

Certificate of Analysis

Report for Targeted Exosomes

Batch Number

Manufacturing Date

Expiry Date

Dosage Form Clear-colourless solution

Storage Temperature 4 °C

TEST

RESULTS

Sterility Test

QUALITY ASSURANCE
PASSED 

Limulus Amoebocyte Lysate Test

QUALITY ASSURANCE
PASSED 

Mycoplasma

QUALITY ASSURANCE
PASSED 

Immunophenotyping

QUALITY ASSURANCE
PASSED 

Nanoparticle Tracking Analysis (NTA)

QUALITY ASSURANCE
PASSED 

Next Generation Sequencing (NGS)

QUALITY ASSURANCE
PASSED 

This certifies that the exosomes delivered by Malaysian Genomics Resource Centre Berhad is deemed suitable for human use as it has been tested, verified and approved in the tests above.

Storage recommendation:

- 1) Refrigerate the vials in 4°C at all times upon receipt
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolonged exposure in room condition
- 4) It is recommended to place the vials in -20°C for prolonged storage

Authorised Signature:



Certificate of Analysis

Batch Number

Manufacturing Date

Expiry Date

Dosage Form

Clear-colourless solution

Storage Temperature

4 °C

TEST		SPECIFICATIONS	TEST RESULTS	REMARKS
1.	Sterility	No growth (no turbidity) in the test media	Non-turbid	QC passed
2.	Limulus Amoebocyte Lysate Test	Endotoxin sensitivity at 0.125	Yes	QC passed
3.	Mycoplasma	Mean of absorbance <0.05	0.03	QC passed
4.	Immunophenotyping	Marker – CD63, CD9, CD81 (≥ 85%).	Within the range	QC passed
5.	Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	9 x 10 ⁹ particles/mL	QC passed
6.	Next Generation Sequencing (NGS)	Exosomal miRNA	hsa-miR-205, hsa-miR-155, hsa-miR-99, hsa-miR-146, hsa-miR-218, hsa-miR-4530	QC passed

Authorised Signature:

Supporting Information

STERILITY TEST ^{5,6}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

MYCOPLASMA TEST ^{4,5}

Negative findings for mycoplasma species *M. hyorhina*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

LIMULUS AMOEBOCYTE LYSATE TEST ^{3,5}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

IMMUNOPHENOTYPING ^{1,2}

Antibodies	Results on Exosomes
CD9	+
CD63	+
CD81	+

* '+' indicates the presence of the marker in the cell.

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.

NEXT GENERATION SEQUENCING (NGS)⁹

miRNA	Function	Regulation
hsa-miR-205	Migration and proliferation of epithelial cells ¹⁰	Up-regulated
hsa-miR-155	Competitive fitness and proliferative potential of Foxp3, regulatory T cells ¹¹	Up-regulated
hsa-miR-99	Cell proliferation, cell migration ¹¹	
hsa-miR-146	Formation of new blood vessels ¹²	Up-regulated
hsa-miR-218	Neovascularization ¹³	Down-regulated
hsa-miR-4530	Angiogenesis ¹⁴	Up-regulated

REFERENCES

- Chettimada, Sukrutha, Lorenz, David R., Misra, Vikas, Dillon, Simon T., Reeves, R. Keith, Manickam, Cordelia, 2018, Exosome markers associated with immune activation and oxidative stress in HIV patients on antiretroviral therapy.
- Joana Maia, Silvia Batista, Nuno Couto, Ana C. Gregório, Cristian Bodo, Julia Elzanowska, Maria Carolina Strano Moraes, Bruno Costa-Silva, 2020, Employing Flow Cytometry to Extracellular Vesicles Sample Microvolume Analysis and Quality Control.
- Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://gmpua.com/Validation/Method/LAL/EUPHARMACOPOEIA.pdf>
- Mycoplasma (USP 63): <http://www.triphasepharmasolutions.com/Private/USP%2063%20Mycoplasma%20Tests.pdf>
- ICH Harmonised Tripartite Guideline, Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q6B.
- USP 71 <Sterility Test>. www.usp.org/sites/default/files/usp/document/harmonization/gen-method/q11_pf_ira_34_6_2008.pdf
- Neha Chopra, Brahm Dutt Arya, Namrata Jain, Poonam Yadav, Saima Wajid, Surinder P. Singh, Sangeeta Choudhury, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
- Dae Hyun Ha, Hyun-keun Kim, Joon Lee, Hyuck Hoon Kwon, Gyeong-Hun Park, Steve Hoseong Yang, Jae Yoon Jung, Hosung Choi, Jun Ho Lee, Sumi Sung, Yong Weon Yi, Byong Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.
- Gautam Singhvia, Prachi Manchanda, Vamshi Krishna Rapallia, Sunil Kumar Dubeya, Gaurav Guptab, Kamal Dua, 2018, MicroRNAs as biological regulators in skin disorders.
- J. Yu, H. Peng, Q. Ruan, A. Fatima, S. Getsios, R.M. Lavker, 2017, MicroRNA-205 promotes keratinocyte migration via the lipid phosphatase, 2010.
- E.J. Mulholland, N. Dunne, H.O. McCarthy, 2017, MicroRNA as therapeutic targets for chronic wound healing.
- H.H. Seo, S.Y. Lee, C.Y. Lee, R. Kim, P. Kim, S. Oh, H. Lee, M.Y. Lee, J. Kim, L.K. Kim, K.C. Hwang, W. Chang, 2017, Exogenous miRNA-146a enhances the therapeutic efficacy of human mesenchymal stem cells by increasing vascular endothelial growth factor secretion in the ischemia/reperfusion-injured heart.
- X. Zhang, J. Dong, Y. He, M. Zhao, Z. Liu, N. Wang, M. Jiang, Z. Zhang, G. Liu, H. Liu, Y. Nie, D. Fan, J. Tie, 2017, miR-218 inhibited tumor angiogenesis by targeting ROBO1 in gastric cancer.
- T. Zhang, L. Jing, H. Li, L. Ding, D. Ai, J. Lyu, L. Zhong, 2017, MicroRNA-4530 promotes angiogenesis by targeting VASH1 in breast carcinoma cells.